

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 06

Miscellaneous Topics



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Recap of Previous Lecture



State transition Diagram

Parity generator & checker





Topics to be Covered

Functionally complete function

Concept of delay.

Functionally complete function.



- T_0 [Preserving all '0's boolean function]:

- $f(A, B) = \overline{A} + \overline{B}$

$$f(0, 0) = 1 \quad \text{Not } T_0$$

- $f(A, B) = A + B$

$$f(0, 0) = 0 \rightarrow T_0$$

$$f(A, B, C, \dots, D)$$

$$f(0, 0, 0, \dots, 0) = 0 \rightarrow T_0$$

$$1 \rightarrow \text{Not } T_0$$

- $f(A, \overline{B}) = A + B$

$$f(0, 0) = 0 + 1 = 1 \quad \text{Not } T_0$$

$$f(\overline{A}, B) = \overline{A} + B$$

$$f(0, 0) = 0 + 0 = 0 \quad T_0$$

T_1 [Preserving all 1's boolean function]

$$f(A, B, C, \dots, D)$$

$$\text{if } f(1, 1, 1, \dots, 1) = 1 \longrightarrow \text{then } T_1$$

$$= 0 \longrightarrow \text{then Not } T_1$$

- $f(A, B) = \overline{A} + B$

$$f(1, 1) = 0 + 1 = 1 \longrightarrow T_1$$

- $f(A, B) = \overline{A} + \overline{B}$

$$f(1, 1) = 0 + 0 = 0 \longrightarrow \text{Not } T_1$$

- $f(A, \overline{B}) = \overline{A} + B$

$$f(1, 1) = 0 + 0 = 0 \quad \text{Not } T_1$$

- $f(A, \overline{B}) = \overline{A} + \overline{B}$

$$f(1, 1) = 0 + 1 = 1 \longrightarrow T_1$$

- $f(A, C) = (\overline{A} + C)$

- $f(A, B, C) = \sum(0, 1, 4, 5, 6)$

↓
Not T_0 , Not T_1

- $f(A, B, C) = \overline{A} + \overline{B} \overline{C}$

↓
Not T_0 , Not T_1

- $f(A, B, C) = \sum(1, 2, 4, 7)$

↓
 T_0 , T_1

- $f(A, B, C) = \sum(3, 5, 6, 7)$

↓
 T_0 , T_1

Linear function:

A function is said to be linear if for all the combinations containing even no. of 1's o/p is '0' and for all combinations containing odd no. of 1's o/p is '1' and vice-versa.

A	B	NL $Y_1 (A+B)$	NL $Y_2 (A \cdot B)$	NL $Y_3 (\overline{A \cdot B})$	NL $Y_4 (\overline{A+B})$	L $Y_5 (A \oplus B)$	L $Y_6 (\overline{A \oplus B})$
0	0	0	0	1	1	0	1
0	1	1	0	1	0	1	0
1	0	1	0	1	0	1	0
1	1	1	1	0	0	0	1

$$f(A, B, C) = \sum(0, 4, 5, 6)$$

NL

$$f(A, B, C) = \sum(0, 3, 5, 6) = \overline{\sum(1, 2, 4, 7)}$$



L

$$f(A, B, \overline{C}) = A + \overline{B} C$$

$$f(A, B, D) = A + \overline{B} \overline{D} \rightarrow$$

NoT T₀, T₁,

NoT L

$$= \sum(0, 4, 5, 6, 7)$$

A		
1	0	0
1	0	1
1	1	0
1	1	1

$\overline{B}$	$\overline{D}$	
0	0	0
1	0	0

- Monotonic function M

→ If any one of i/p changes from 0 to 1 and function does not decrease then it will be said as monotonic function.

A	B	M $y_1(A+B)$	M $y_2(A \cdot B)$	NM $y_3(\overline{A \cdot B})$	NM $y_4(\overline{A+B})$	NM $y_5(A \oplus B)$	NM $y_6(\overline{A \oplus B})$
0	0	0	0	1	1	0	1
0	1	1	0	1	0	1	0
1	0	1	0	1	0	1	0
1	1	1	1	0	0	0	1

$S_D \rightarrow$ self dual



- Functionally complete function:

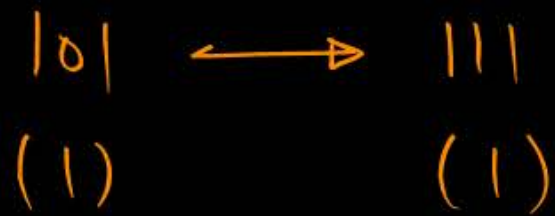
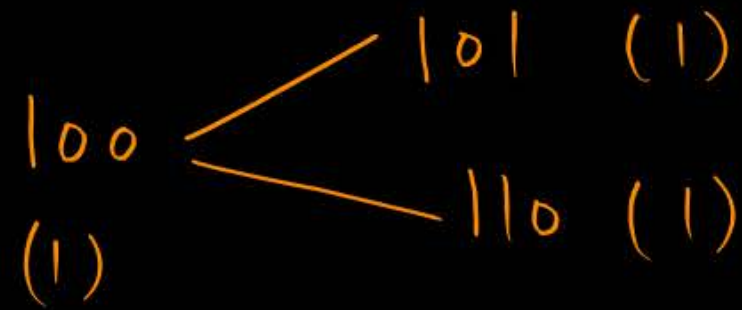
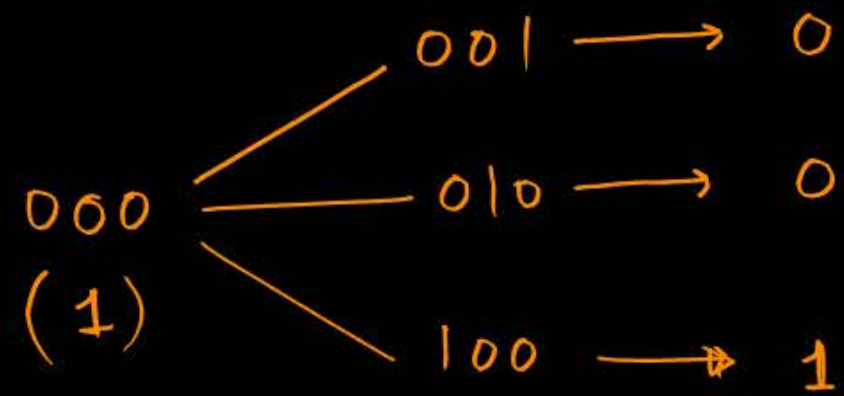
A function is said to be functionally complete if it does not belong to any of the category T_0, T_1, L, M & S_D .



#Q. $f(A, B, C) = A + \overline{B}\overline{C} = \Sigma(0, 4, 5, 6, 7)$

NoT T₀, T₁, NL, NSD, NM

Not functionally complete



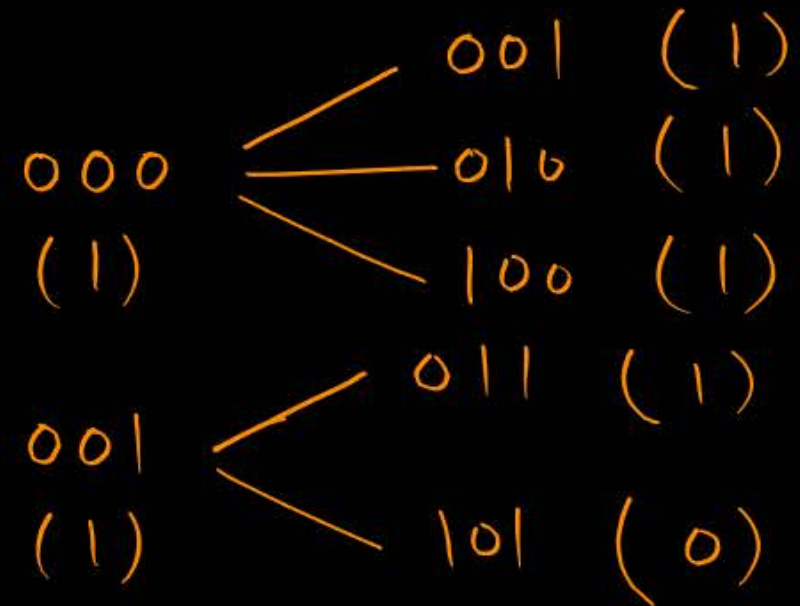


#Q. $f(A, B, \bar{C}) = \bar{A} + \bar{B}C$

$$f(A, B, D) = \bar{A} + \bar{B}\bar{D} = \sum(0, 1, 2, 3, 4)$$

Not T_0 , Not T_1 , NL, NSD, NM

functionally complete
function.





#Q. $f(A,B,C) = AB + BC + CA = \sum(3,5,6,7)$

T_0, T_1, NL, S_D, M

Not functionally
complete.

$$\begin{array}{ccc} 011 & \longrightarrow & (111) \\ (1) & & (1) \end{array}$$

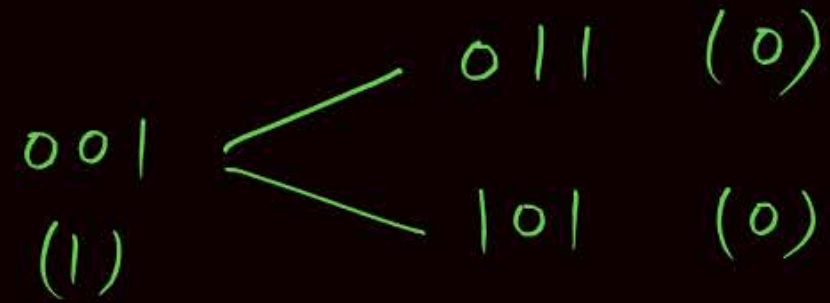
$$\begin{array}{ccc} |0| & \longrightarrow & (111) \\ (1) & & 1 \end{array}$$

$$\begin{array}{ccc} ||0 & \longrightarrow & 111 \\ (1) & & (1) \end{array}$$

Q. $f(A, B, C) = A \oplus B \oplus C = \sum(1, 2, 4, 7)$

$T_0, T_1, L, S_D, NM.$

Not functionally
complete.



#Q. $f(A, B) = (\bar{A} + B) = \sum(0, 1, 3)$

NOT, T₀, T₁, NL, NSD, NM.

Not functionally complete

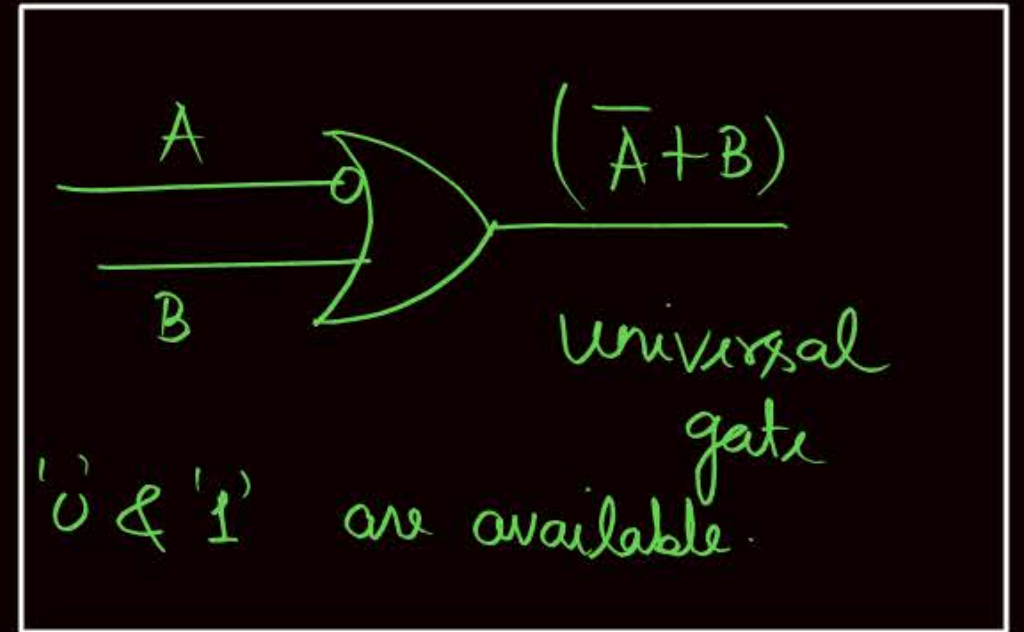
$$f(A, A) = \bar{A} + A = 1$$

$$\begin{array}{cc} 00 & \begin{array}{l} \swarrow \\ \searrow \end{array} \begin{array}{l} 01 \quad (1) \\ 10 \quad (0) \end{array} \\ (1) & \end{array}$$

→ NOT is not possible

↓
functionally not complete function.

Here '0', & '1' are not available by default.



#Q. $f(A, B) = \bar{A}B + A\bar{B} = \Sigma(1, 2)$

$T_0, \text{NOT } T_1, L, NSD, NM.$

functionally
incomplete.

01	——	11
(1)		(0)

$$\Rightarrow \{f_1, f_2, \dots, f_3, \dots\}$$

A set of boolean function is said to be functionally complete if

• There is atleast one function which does not belong to T_0 .

•)))))))))))))))))) to T_1 .

•)))))))))))))))))) L.

•)))))))))))))))))) M.

•)))))))))))))))))) S_D .

Q. $f = \{ f_1(A, B) = A \oplus B, \quad f_2(A, B) = (A+B) \}$

T_0

L

$\text{NOT } T_1$

$\text{NOT } M$

$\text{NOT } S_D$

T_0

$\text{NOT } L$

T_1

M

$\text{NOT } S_D$

Not functionally complete.

$$\begin{array}{lcl} 01 & \longrightarrow & 11 \\ (1) & & (1) \\ 10 & \longrightarrow & 11 \\ (1) & & (1) \end{array}$$

$$\# f = \{ f_1(A, B) = A \oplus B, \quad f_2(A, B) = (A+B), \quad f_3(A) = \overline{A} \}$$

T_0

L

NoT T_1

NoT M

NoT S_D

T_0

NoT L

T_1

M

NoT S_D

NoT T_0

L

NoT T_1

MM

S_D

$\Sigma(0)$

$$\begin{array}{ccc} 0 & \longrightarrow & 1 \\ (1) & & (0) \end{array}$$

functionally complete set.

$$f_3^D(A) = \overline{A} = f_3(A)$$

Implication function

$$P \longrightarrow Q$$

NOT T_0 , T_1

NOT L , NOT S_D

NM

NOT functionally complete

$$Q \rightarrow P$$

$$f(P, Q) = (\overline{Q} + P)$$

P	Q	$y[P \rightarrow Q]$
0	0	1
0	1	1
1	0	0
1	1	1

$$y[P, Q] = \Sigma(0, 1, 3) = \pi(2) \\ = (\overline{P} + Q)$$

$$\begin{array}{l} 00 \\ (1) \end{array} \begin{array}{l} \text{---} 01 (1) \\ \text{---} 10 (0) \end{array}$$

Negation

$$f(A) = \overline{A} = \Sigma(0)$$

NoT T₀
 NoT T₁
 L
 S_D
 NM

0 → 1
 (1) (0)

$$f = \left\{ \begin{array}{l} \text{Implication, Negation} \\ f(A, B) = \overline{A} + B, \quad f(A, \overline{A}) \end{array} \right\}$$

A → B



T₁
 NoT T₀
 NoT L
 NoT S_D
 NoT M

NoT T₀

NoT T₁

L

S_D

NM.

Overall set is functionally complete.

H.W

Q.1. $f(A, B, C) = \overline{A} + \overline{B}C$, T_0, T_1, L, M, S_D

Q.2. $f(A, \overline{B}) = \overline{A}B$, T_0, T_1, L, M, S_D

Q.3. $f(\overline{A}, B) = A + \overline{B}$, T_0, T_1, L, M, S_D

Q.4. $f(\overline{A}, \overline{B}) = \overline{A}B$, T_0, T_1, L, M, S_D

Q.5. $f(A, \overline{B}, C) = \overline{A}B + \overline{C}$, T_0, T_1, L, M, S_D

$$f(A, \overline{B}) = \overline{A} + \overline{B}$$

00	→
01	→
10	→
11	→

$$f(A, \bar{B}) = \bar{A}B$$

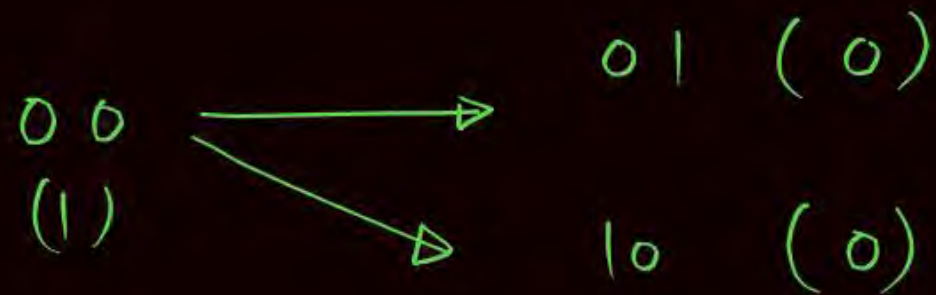
$$f(A, C) = \bar{A}\bar{C} = \Sigma(0)$$

NoT T_0 , NoT T_1

NoT L, NoT S_D

NM

functionally complete
function.





Topic : 2 Min Summary

functionally complete
function.

Thank you

GW
Soldiers !

